Orthographic Variation and Input Normalization in Persian LLM Question Answering: A Paired Evaluation

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***Abstract*—Persian allows several spellings of the same word: the Arabic and Persian forms of yeh and kaf look identical on screen, and a word boundary may be a space, a zero-width non-joiner (ZWNJ) or nothing at all. Readers tolerate these variants, whereas a tokenizer processes them as different strings. We ask each multiple-choice question under six conditions: as written (C), with a controlled spelling change (P(C)), and after each of two normalizers. N is a deterministic inverse-map baseline, and H is Hazm 0.10.0. We approved the items by joint review, which left 97 yeh/kaf items and 69 ZWNJ-to-space items. Qwen2.5-1.5B-Instruct (Q4_K_M) answered three requests per eligible condition, 2,982 in all, and every response could be scored. On yeh/kaf, C and P(C) each got 138/291 requests and 46/97 item majorities right. Seven items got worse and seven got better (exact McNemar $p = 1.0$), and the request-level difference was 0.0 points (95% bootstrap CI -7.2 to 7.2). On ZWNJ, C got 94/207 and P(C) 96/207, and both got 32/69 majorities. Two items got worse and two better ($p = 1.0$), and the difference was $+1.0$ point (CI -3.9 to 5.8). N recovered 21 of 23 yeh/kaf request-level losses, all by restoring C exactly. H recovered 17 of 23, five of them without restoring C exactly. On ZWNJ, N recovered 1 of 5 without exact restoration, and H recovered 4 of 5 by exact restoration. Identical inputs still got different answers now and then. We saw no aggregate accuracy difference; because the intervals allow modest effects either way, this is no evidence of equivalence. A frozen targeted replication then sent 180 requests to Qwen2.5-7B-Instruct, covering 20 yeh/kaf and 10 ZWNJ pairs. On those 30 items, 1.5B C got 43/90 requests (15/30 item majorities) and 7B C got 21/90 (7/30). Of the C-correct requests, 23/43 (53.5%) became P(C)-incorrect for 1.5B and 0/21 (0%) for 7B. Because the 7B sample had low absolute accuracy, this disappearance is inconclusive and cannot be attributed to greater robustness.**

***Index Terms*—Persian NLP, orthographic variation, ZWNJ, Hazm, input normalization, multiple-choice question answering, robustness, cross-model replication.**

I. Introduction

A Persian question has no single canonical spelling: the Arabic and Persian forms of yeh (ي, ى) and kaf (ك, ک) are different code points that most readers cannot tell apart, and word boundaries are just as loose, since an ordinary space, a ZWNJ or no visible gap can all occur in the same position. Readers are largely unaffected by this variation; the tokenizer, however, receives different strings and the model different token sequences.

Small prompt changes are known to move model answers. Romanou et al. [1] show this on several established benchmarks and separate how hard an item is from how sensitive it is to its surface form. Aliakbarzadeh et al. [2] build noisy multilingual test sets from Wikipedia edit histories; Persian is not among their languages. For Persian itself, normalization has been studied in sentiment classification [3], colloquial-to-standard rewriting [4] and retrieval [5]. None of these follows answer changes through paired Persian multiple-choice questions, and that is the gap this study addresses.

We ask whether a spelling change that a reader would not notice alters the model's answer, and whether a fixed normalizer then restores it or breaks an answer that was correct. Normalization that simply returns the text to its original bytes will look like a perfect fix, so those cases are reported separately.

This paper makes three contributions. First, it specifies a paired protocol in which each question is asked in six conditions with fixed choices and a fixed instruction, together with metrics that keep sensitivity, recovery, harm, exact restoration and identical-input variability apart. Second, it reports an exploratory evaluation of a small quantized model on 97 yeh/kaf and 69 ZWNJ items approved by joint author review. Third, it adds a frozen, targeted replication with a 7B model and states plainly what that replication can and cannot show.

The rest of the paper is organized as follows. Section II reviews related work, and Sections III and IV state the research questions and define the metrics. Section V describes the data and item screening, Section VI the experimental design and Section VII the analysis plan. Results are reported in Sections VIII and IX, followed by a discussion, the limitations and the conclusion in Sections X–XII.

II. Related Work

The cited studies examine different interventions, so their numbers are not pooled; we group them by their relevance to this study.

A. Prompt sensitivity and noise

Brittlebench [1] applies word-level, formatting and paraphrase changes to several benchmarks and compares each item with its own perturbed versions. We adopt that within-item logic directly. Brittlebench includes neither Persian Unicode variants nor a step in which a normalizer repairs the damage. Aliakbarzadeh et al. [2] draw noisy test sets for NLI, NER and intent classification in six languages. Their comparison of

clean and noisy versions covers no Persian, no multiple-choice QA and no normalization.

B. Robustness beyond spelling

Bogavelli et al. [6] test punctuation, whitespace, structured formats, multilingual inputs and instruction position in enterprise settings, and report losses of up to 40 percentage points on some metrics across 11 models. Their figure is specific to their setting, yet it gives a reason to measure "minor" edits instead of assuming they are harmless. Nhayakkat et al. [7] reorder constituents and flip active and passive in Hindi and Malayalam math problems, and treat meaning preservation as something the experiment has to check. ObfusQAte [8] uses entity indirection, distractors and contextual overload on factual QA. Although none of this concerns Unicode normalization, it shows why a robustness study should state exactly which perturbations it uses.

Sperduti and Moreo [9] survey the wider misspelling literature. They split noise into natural, artificial, hybrid and adversarial, note that no shared robustness benchmark exists, and review evidence that synthetic misspellings can behave differently from real errors. We therefore report the exact character map, the edit positions and the number of changed characters rather than calling everything "noise".

C. Persian orthography and normalization

Khazeni et al. [3] describe a slang-to-formal pipeline whose preprocessing unifies Persian and Arabic letters and fixes spaces and semi-spaces. Ghayoomi et al. [4] test five web LLMs on colloquial-to-standard rewriting. Naebzadeh et al. [10] survey the informal-to-formal area and point to gaps in length balance and dialect coverage. Zoughi et al. [5] handle character and whitespace normalization inside a Persian RAG system. They all show the problem is real. What they measure is rewriting quality or downstream classification, not whether an answer survives a spelling change.

Dashti et al. [11] bring in context. Their spelling corrector, PERCORE, combines contextual representations with phonetic analysis and scores real-word and non-word errors separately. A character map can be exactly reversible in Unicode and still alter the meaning of a particular question, so we screen out spelling-sensitive items and review each ZWNJ edit individually.

Abdullah et al. [12] train Arabic-script RoBERTa models separately for Kurdish Sorani, Arabic, Persian and Urdu, and report gains of 2–5 percentage points over multilingual baselines on language-specific classification. That supports handling shared-script variation language by language; it does not tell us what an off-the-shelf LLM does with a yeh or kaf swap. Two more papers mark the edge of the term: Chen et al. [13] study normalization inside transformer architectures, and Sunkara et al. [14] study inverse text normalization for speech recognition. Neither deals with Persian input.

D. Benchmark quality and human review

The FarsEval-PKBETS paper [15] describes a 4,000-question Persian benchmark with two main reviewers who supervised question quality and sent items back for revision. It is our best example of why human verification matters. We have not matched our five local JSON files to that release, so we cite the paper as a benchmark source and keep the files under their repository name [16].

Other benchmarks make the same point. MedHopQA [17] has 1,000 expert-curated biomedical questions validated through a human–AI pipeline. ToolQA [18] leans on automated curation and tool-based verification. PerSoMed [19] pairs ChatGPT few-shot annotation with human checking and gets weaker results on the more ambiguous classes. None of this validates our items, although it shows that the reviewer, the protocol and the disagreements should all be written down.

Liu [20] compares gpt-4o-mini labels with three crowdworker labels on a rubric task and finds that disagreements cluster around adjacent severity levels. Although it is a subjective-labeling study rather than factual MCQ validation, it supports our decision to leave the human-verification fields empty after the AI-assisted first pass. Mortazavi et al. [21] call Persian annotation-scarce. A dataset can hold many records and still lack the item-level evidence a causal claim needs, so the 144/8/5 split below is data preparation, not a quality assessment.

E. Persian models and resources

Hakim [22] is a Persian embedding model released with new retrieval datasets. Persian-Phi [23] adapts the compact 3.8B Phi model, and PersianLLaMA [24] reports 7B and 13B Persian models. Although they are relevant to model choice on a small budget, none of them runs an experiment on orthographic pairs. ROBUSTFT [25] deals with noisy responses during fine-tuning, which is a training-data problem, not input noise. The Persian proverb study [26] shows how hard cultural meaning stays even when the text is fluent, a reason to keep culture-sensitive questions visible in the review record.

F. Positioning

Prompt sensitivity, structural perturbation, misspelling robustness, Persian standardization, Persian benchmarks and human–AI annotation are all covered somewhere. What nobody has run, as far as our collection shows, is this: on author-approved Persian MCQs, do reversible Arabic/Persian character or spacing changes alter answer correctness, and do fixed normalization procedures bring the reference answer back without adding errors? This study addresses that gap.

G. Evidence matrix

Table I summarizes the closest sources. The reviewed literature covers adjacent problems, but no single source combines Persian MCQs, reversible orthographic perturbations, a normalization-recovery arm and repeated identical-input controls.

Prior work separates into (a) general prompt and noise robustness, (b) Persian rewriting or spelling correction, and (c) Persian benchmark construction. These lines differ in

TABLE I
EVIDENCE MATRIX OF THE CLOSEST RELATED WORK.

Reference	Technical approach	Application / context	Evaluation criteria	Advantage	Disadvantage / unresolved issue	Relation to this study
Romanou et al. [1]	Within-item prompt perturbations across benchmark tasks	General LLM robustness; MCQ included	Accuracy and sensitivity under paired changes; quality checks for paraphrases	Closest general paired-robustness design	No Persian Unicode variants, normalization recovery or identical-input repeat analysis	We transfer the paired logic to Persian orthography
Aliakbarzadeh et al. [2]	Wikipedia-edit and synthetic noisy test sets	Multilingual NLI, NER, intent classification	Clean/noisy performance under fixed noise ratios	Real-world noise provenance; cross-language comparison	Persian absent; no normalization arm or QA answer preservation	Motivates explicit perturbation provenance
Khazeni et al. [3]	Rule-based and statistical slang-to-formal conversion; letter and spacing cleanup	Persian social-media sentiment	Conversion quality and downstream classification	Direct Persian normalization precedent	Unpaired rewriting task; no MCQ recovery or harm metric	Supports the language-specific motivation
Ghayoomi et al. [4]	Five web LLMs convert colloquial Persian to standard writing	Persian orthography and rewriting	Expert categories against an orthography guide	Close language and orthography context	Output standardization, not controlled input perturbation; metadata conflicts remain	Shows that meaning preservation must be checked separately
Zoughi et al. [5]	Character and whitespace normalization in a hybrid Persian RAG system	Persian retrieval-augmented generation	Retrieval and generation metrics across configurations	Practical normalization and segmentation relevance	Unpaired system comparison; unclear denominators; no QA robustness control	Supports logging exact changes
Sperduti and Moreo [9]	Survey taxonomy of natural, artificial, hybrid and adversarial spelling noise	Cross-task NLP robustness	Taxonomy and synthesis of reported metrics	Warns that synthetic and natural noise differ	Does not prescribe a Persian MCQ recovery experiment	Defines perturbation terminology
FarsEval-PKBETS [15]	Large Persian benchmark with several response formats and a review workflow	Persian knowledge, medicine, law, culture and generation	Benchmark accuracy and reviewer-supervised item revision	Strong context for Persian benchmark quality	Identity with the local JSON release is unresolved; no orthographic perturbation arm	Sets a verification standard without being treated as our dataset
PERCORE [11]	Contextual Persian spelling correction with phonetic analysis	Persian real-word and non-word correction	Correction precision and recall-style scores	Shows that context can make normalization unsafe	Correction scores do not measure downstream MCQ answer recovery	Motivates separate harm and exact-restoration metrics

language, task, perturbation source and outcome. The missing combination is a paired Persian MCQ design that keeps the choices fixed, verifies meaning preservation, sends clean and perturbed text to the same model, tests a normalizer, and reports recovery, harm, exact restoration and identical-input variability separately.

III. Research Questions

RQ1 (Sensitivity). Among author-approved Persian MCQs, how often does a controlled, meaning-preserving orthographic variant turn a correct answer into an incorrect one?

RQ2 (Recovery and harm). Among items answered correctly in C and incorrectly in P(C), how often are N(P(C)) and H(P(C)) correct? How often does either normalizer turn a correct answer into a wrong one?

RQ3 (Interpretation and repeatability). How do exact-restoration cases differ from the rest, and how often do repeated identical inputs get different labels?

IV. Formal Definitions and Estimands

The literature motivates the design without fixing one universal metric, so we define the estimands before looking at any model output.

A. Notation and metrics

For item i, condition x and repeat r, let $Y(i, x, r) = 1$ if the parsed label equals the verified answer key, and $Y(i, x, r) = 0$ otherwise. Let N_x be the number of scorable requests in condition x, and n the number of items.

- *Request accuracy:* the number of requests with $Y = 1$, divided by N_x.
- *Item-majority correctness:* $M_i(x) = 1$ when at least two of the three repeats are correct. Item-majority accuracy is the number of items with $M_i(x) = 1$, divided by n.
- *Perturbation loss* L: the number of (item, repeat) pairs with $Y(i, C, r) = 1$ and $Y(i, P(C), r) = 0$. The *conditional discordance rate* is L divided by the number of C-correct requests; when that denominator is zero the rate is undefined, not zero.
- *Recovery* for a normalizer G: the fraction of the L losses for which the answer to $G(P(C))$ is correct.
- *Clean-input harm:* among C-correct requests, the fraction for which $G(C)$ is answered incorrectly. *Perturbed-input harm* is the same fraction among P(C)-correct requests, with $G(P(C))$ in place of $G(C)$.
- *Exact restoration:* a text predicate, $G(P(C)) = C$. It is reported separately from answer recovery, because recovery can occur without restoring the original bytes.

B. Inference and uncertainty

The primary paired endpoint is one item-majority outcome per item for C versus P(C). A two-sided exact McNemar test is used only for this paired binary endpoint. Request-level differences are summarized with an item-clustered percentile bootstrap (10,000 resamples, fixed seed), which keeps all repeats of an item together. Repeats are not treated as independent questions.

C. Prespecified algorithm

1) Load the JSON arrays or JSONL files and validate that each answer index matches its answer text.

2) Screen and author-review candidate MCQs, excluding ambiguous or spelling-sensitive items.
3) Generate C, P(C), N(C), N(P(C)), H(C) and H(P(C)), recording edit positions, counts, hashes and exact-restoration flags.
4) Send each condition in an independent local request with the fixed Persian instruction and an A–D grammar; cap retries and log transport errors separately.
5) Parse one label, align results by item, condition and repeat, and compute request and majority outcomes, loss, recovery, harm, exact restoration and identical-input disagreement.
6) Report family-specific and model-specific results with denominators, confidence intervals and open limitations, without pooling incompatible perturbation families.

Interpretation boundary. If N(P(C)) equals C, a recovery can be a restoration effect and not evidence for a new method. If N(C) equals C, differences between independent model calls are response variability, not clean-input harm caused by text normalization.

V. Data and Item Screening

A. The local files

The five local files come from the main/data directory of Persian LLM Eval [16]; we did not record the exact commit. They hold 554 records: 100 challenge, 10 development, 145 hard, 150 practical and 149 public-evaluation. No identifier appears twice. Of these, 157 are marked as multiple-choice, and in every one of them the answer index points to the same text as the stored answer. The metadata tag 250 records as *pending_review* and 254 as *accepted*, and 50 carry no review status at all. Those tags describe the files; they do not tell us anyone checked the answers.

The two of us, both Persian speakers, reviewed the 40 pilot questions, their keyed answers and whether meaning survived in all four conditions. We worked independently first and settled disagreements together. This is human verification done by the authors, not an outside review. We agreed that every yeh/kaf variant kept the intended meaning. We excluded `peval-hard-reasoning-002` and `-024`, and classed `-005`, `-007` and `-012` as ambiguous and left them out of the cleaned analysis.

After the ZWNJ run, we reviewed all 24 ZWNJ-eligible questions, their choices, stored keys and transformed texts, again independently, and our decisions were identical. Twenty items were approved for a post hoc reviewed analysis. We rejected reasoning-005 and reasoning-024 for key or inference defects, excluded reasoning-009 as uncertain because its conditional states an intention and not a guaranteed rule, and kept reasoning-007 out under the earlier ambiguity decision. We also checked that all 11 clean-input edits made by Hazm preserved meaning and the stored answer. As before, this is author-conducted review, not external expert validation.

On 21 October 2026 we jointly reviewed the full text and four choices of the 117 structurally consistent MCQs outside the 40-item pilot. This was joint adjudication, not independent rating, so we make no kappa or agreement claim. We approved 97 items, marked 11 for rewriting, rejected two semantic duplicates and left seven uncertain. We then jointly reviewed every generated expansion transformation before anything ran. All 97 yeh/kaf P/N branches were approved. Only 96 were approved for all six conditions, because Hazm changed the meaning of `peval-public-knowledge-002`. All six conditions were approved for the 69 ZWNJ-eligible items. These decisions defined the frozen expansion sets.

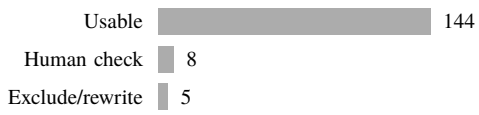


Fig. 1. Preliminary screening of the 157 structurally consistent items. The chart describes data preparation only; it is not a model-accuracy result.

B. Preliminary screening

All 157 questions went through a quick editorial screen done with AI assistance, not by an independent human reviewer. For each item it compared the stem, the choices and the stored answer index, and looked for duplicate answers, ambiguous wording, missing choices, a mismatch between question and choices, and answers that depend on time, region or culture. It did not check facts against outside sources. Fig. 1 shows the outcome: 144 items usable for now, 8 needing a human check, and 5 to exclude or rewrite.

The 144 items showed no obvious competing answer or structural defect, which is a lower bar than having verified facts. The eight held back involve culturally variable expressions, folk beliefs, context-dependent pragmatics or knowledge that changes over time. Five items have a plain defect and stay out as written: two choices give the same answer (`hard-reasoning-013`); the expression جا انداختن is ambiguous without context (`hard-culture-020`); the choices do not match what the stem asks (`public-knowledge-014`, `-017`, `-023`). Table II lists all thirteen flagged items with their status after our later joint review, translated from the original Persian notes. That review did not affect any completed model run. The joint record of the 117 non-pilot items stays under `work/` because it contains third-party text.

The author review covered only the 40 pilot items. The rest of the 144 usable questions and the 8 flagged ones still need the same treatment. The review sheets list record IDs and SHA-256 prompt hashes instead of question text, since we have not settled whether the data can be redistributed.

VI. Experimental Design

A. Six paired conditions

Each question is asked under up to six conditions (Table III, Fig. 2). Choices, labels, order and the response instruction are byte-identical in all of them, and the model never sees the reference answer. The flag `--hazm-normalizer` adds the two Hazm conditions explicitly.

TABLE II

THE THIRTEEN INITIALLY FLAGGED ITEMS AFTER JOINT REVIEW. THE DEFECTS IN PUBLIC-KNOWLEDGE-014 AND -017 ARE STRUCTURAL, BECAUSE NO OPTION CAN FULLY ANSWER THE STEM. WE KEEP "REWRITE" AND "UNCERTAIN" AS SEPARATE CATEGORIES.

Record ID	Status	Rationale	Example or implication
`peval-hard-reasoning-013`	Rewrite	Two choices express effectively the same answer.	Joint answer was A|D; replace option D.
`peval-hard-culture-020`	Rewrite	جا انداختن has several meanings without enough context.	Add explicit classroom context or remove the item.
`peval-public-knowledge-014`	Rewrite	The stem asks for an element and group, but the choices give only names.	Ask only for the element, or add groups.
`peval-public-knowledge-017`	Rewrite	The HTTPS/TLS answer is absent.	Recorded NONE.
`peval-public-knowledge-023`	Rewrite	The stem and the answer type do not match.	Recorded NONE.
`peval-hard-culture-001`	Uncertain	The idiom depends on context, and the keyed gloss is broad.	Needs review by people from more varied backgrounds.
`peval-hard-culture-002`	Uncertain	Rejecting taarof depends on context.	The keyed choice is plausible but weakly constrained.
`peval-hard-culture-009`	Rewrite	Calling the utterance "polite" is contestable.	Use "a direct request not to interfere."
`peval-hard-culture-015`	Rewrite	No universal repetition count exists for taarof.	Remove the arbitrary "one or two" wording.
`peval-practical-pragmatics-006`	Uncertain	صبر آمد varies by region and generation.	Needs review by people from more varied backgrounds.
`peval-public-knowledge-005`	Rewrite	The population ranking changes over time.	Add a census year and recheck the key.
`peval-public-culture-016`	Rewrite	The keyed choice combines two occasions.	Separate the occasions or narrow the stem.
`peval-public-culture-019`	Uncertain	The expression depends on context.	Drinking is plausible, but "most often" is contestable.

TABLE III

THE SIX PAIRED CONDITIONS.

Condition	Description	Model-visible change
C	Verified reference	None
P(C)	Controlled variant	Question body only
N(C)	C after our rule-based normalizer	Custom rules
N(P(C))	P(C) after the same normalizer	Custom rules
H(C)	C after Hazm 0.10.0	Off-the-shelf
H(P(C))	P(C) after Hazm 0.10.0	Off-the-shelf

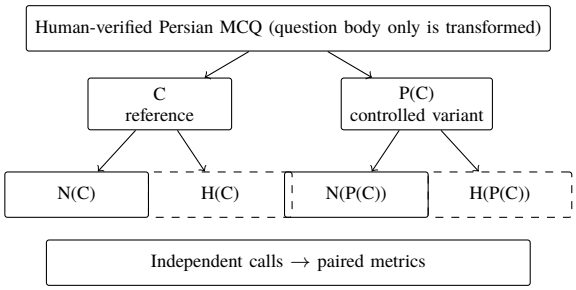


Fig. 2. Study design. Each question branches into six conditions, each is sent in an independent call, and the results are analyzed as pairs. Dashed boxes denote the Hazm branches.

B. Perturbations

The character perturbation swaps Persian yeh and kaf for their Arabic counterparts: ی → ي and ک → ك. N applies that reverse map explicitly, plus only the literal spacing rules. The spacing perturbation replaces an existing ZWNJ with an ordinary space and logs every edit. Its rules live in `code/spacing_rules.json`, and nothing deletes spaces globally. If a transformation leaves the string unchanged, the variant is discarded.

Although ZWNJ edits look harmless in most words, they can change how a tokenizer splits a word and occasionally what the word means, so each one gets an item-level review. The yeh/kaf swap differs in that it can be undone exactly, so N(P(C)) is often byte-identical to C. We analyze those exact-restoration cases separately so they do not inflate the recovery rate.

Hazm is a more demanding test of the normalization step. H is `hazm.Normalizer` from Hazm 0.10.0 [27] with all eight constructor switches on: `correct_spacing`, `remove_diacritics`, `remove_specials_chars`, `decrease_repeated_chars`, `persian_style`, `persian_numbers`, `unicodes_replacement` and `seperate_mi`. It does more than undo one character swap, so it can change a clean question as well. Every textual edit it makes is logged, which makes the comparison with N meaningful: a Hazm "recovery" is no longer just the original string coming back.

C. Response format

Every condition runs in a fresh context with the same short instruction, which asks for the letter of the correct choice and nothing else:

هیچ توضیح دیگری ننویس فقط حرف گزینهٔ درست را بنویس (مثلاً A)

The code prepares requests offline, imports responses produced elsewhere and resumes a capped run. It calls an API only with explicit authorization and the `--execute` flag.

D. The runs

We ran four local experiments with the same 1.5B model and engine; a fifth, targeted run with a 7B model is described

in Section IX. The model was `Qwen2.5-1.5B-Instruct-Q4_K_M.gguf` (SHA-256 `1adf0b11065d8ad2e8123ea110d1ec956dab4ab038eab665614adba04b6c3370`), served through llama.cpp build 11612 (commit 10da8c463). The model repository declares Apache-2.0 terms [28]. Inference ran on an Intel i5-6500 CPU with four logical processors and about 12 GB of RAM. No paid API was involved, and every condition went to the local server as its own request.

The first pilot, on 17 October 2026, used four conditions on 40 items from the screened pool, one request each. The model ran before any human check: the reference answers came from the AI-assisted screen, and the author review came afterwards. It used temperature 0, seed 20261017, a one-token cap, a grammar limiting output to A–D, and shuffled request order.

The second run, on 18 October 2026, kept the model, engine, decoding settings and shuffle seed, added H(C) and H(P(C)), and sent three independent requests per item and condition. It tested only the yeh/kaf swap, applied to the question body and not to the choices, with the spacing rules switched off. All 720 responses could be scored. Hazm 0.10.0 pins NumPy 1.24.3, which does not work with Python 3.12, so the run used Hazm 0.10.0 with documented dependency overrides.

The third run tested the ZWNJ-to-space change on the 24 eligible items, with the same model settings, all six conditions and three repeats: 432 requests, all scorable. It ran before our detailed two-author review of the ZWNJ variants.

The expansion ran on 21 October 2026 with the same model, engine, prompt, decoding constraints, seed and three repeats. For yeh/kaf it sent 1,164 requests for C, P(C), N(C) and N(P(C)) on 97 items, and 576 requests on the 96-item subset that was safe for Hazm. For ZWNJ it sent 1,242 requests across all six conditions on 69 items. All 2,982 responses were scorable.

VII. Analysis Plan

We wrote the analysis down before the six-condition run, in `analysis_plan_v8.md`. It fixed the 35-item set kept by the author review as the primary analysis and all 40 items as a secondary one, with three requests per item and condition. The five exclusions come from our earlier review, so they were settled before this run. The plan named these quantities: request-level and item-majority accuracy, perturbation loss among C-correct observations, N-versus-H recovery, harm on clean and on perturbed inputs, exact recovery separated from non-exact recovery, and disagreement between identical inputs. Anything added beyond that is labelled post hoc.

Before the ZWNJ run we wrote a second plan. It fixed all 24 eligible items as a transparency set and 21 items as the exploratory primary set, dropping three records already flagged for item-quality concerns (`-005`, `-007` and `-024`). The 20-item set that also drops the uncertain `-009` is therefore a post hoc sensitivity analysis and does not replace the frozen 21-item result.

TABLE IV
Yeh/kaf expansion accuracy. The Hazm rows cover the 96-item subset.

Condition	Request accuracy	Item-majority accuracy
C	138/291 (47.4%)	46/97 (47.4%)
P(C)	138/291 (47.4%)	46/97 (47.4%)
N(C)	138/291 (47.4%)	45/97 (46.4%)
N(P(C))	141/291 (48.5%)	48/97 (49.5%)
H(C)	134/288 (46.5%)	45/96 (46.9%)
H(P(C))	133/288 (46.2%)	42/96 (43.8%)

Accuracy is reported for every condition that was run. Whether H(P(C)) equals H(C) is a different question from whether H(P(C)) equals the original C, and we keep the two apart. Transport failures and unscorable outputs are counted separately from wrong answers.

We do not treat N as a rival to H. For yeh/kaf, N is a deterministic inverse-map positive control, and its exact recoveries are expected by construction. H is a general-purpose, off-the-shelf normalizer that may change more than we asked of it. We therefore report their recovery, harm and exact-restoration counts separately and make no claim about which one ranks higher.

For the expansion, the primary inferential endpoint was item-majority correctness for C versus P(C), fixed in advance, tested with a two-sided exact McNemar test with one paired majority outcome per item. For request-level differences between C and P(C), we give percentile 95% confidence intervals from 10,000 item-clustered bootstrap samples (seed 20261021), keeping all repeats of each sampled item together. The earlier pilot runs (40 and 24 items) received no confidence intervals or tests, because their denominators are small and five items were filtered only after the first run. The targeted 7B replication (Section IX) followed a separate frozen plan; its items were chosen from 1.5B outcome strata before any 7B request was sent, so it makes no claim about prevalence.

VIII. Results

The expansion completed all 2,982 requests, and every response could be scored. We do not pool the two perturbation families.

A. Yeh/kaf expansion

For C versus P(C), 39 items were right in both conditions and 44 wrong in both. Seven went from right to wrong and seven from wrong to right. Item-majority accuracy did not move, and the exact McNemar result was $p = 1.0$. At request level the difference was 0.0 points, with a 95% CI from -7.2 to 7.2.

B. ZWNJ expansion

For C versus P(C), 30 items were right in both and 35 wrong in both. Two got worse and two got better. Item-majority accuracy was identical (exact McNemar $p = 1.0$), and the request-level difference P(C)$-$C was $+1.0$ point (95% CI -3.9 to 5.8).

TABLE V
ZWNJ EXPANSION ACCURACY.

Condition	Request accuracy	Item-majority accuracy
C	94/207 (45.4%)	32/69 (46.4%)
P(C)	96/207 (46.4%)	32/69 (46.4%)
N(C)	95/207 (45.9%)	32/69 (46.4%)
N(P(C))	95/207 (45.9%)	32/69 (46.4%)
H(C)	91/207 (44.0%)	30/69 (43.5%)
H(P(C))	94/207 (45.4%)	31/69 (44.9%)

TABLE VI
RECOVERY AND HARM IN THE EXPANSION, BY PERTURBATION FAMILY.

	Yeh/kaf		ZWNJ	
Metric	N	H	N	H
Perturbation loss	23/138		5/94	
Recovery among losses	21/23	17/23	1/5	4/5
Exact recovery	21/23	12/23	0/5	4/5
Non-exact recovery	0/23	5/23	1/5	0/5
Harm, clean input	8/138	8/135	1/94	4/94
Harm, perturbed input	22/138	23/135	5/96	6/96

C. Recovery and harm in the expansion

On yeh/kaf, every N recovery restored C exactly, which is what an inverse map does by construction. H recovered 17 of the 23 losses: 12 by restoring C exactly and five without. N(C) and N(P(C)) were the same string as C in this family, so differences between their independent calls are the model varying, not changed text. Among 214 repeated identical-input groups, 22 disagreed in label and 19 in correctness.

On ZWNJ, N recovered one loss without restoring C, and H recovered four by restoring C exactly. Sixteen of 174 repeated identical-input groups disagreed in label and ten in correctness. These counts do not support a general ZWNJ accuracy effect or a general benefit from either normalizer.

D. Earlier yeh/kaf run (40 items)

The first six-condition run completed 720 of 720 requests. Its 35-item analysis was fixed in advance as primary, and all 40 items as secondary. The similarly filtered Version 7 result is post hoc and is not pooled here.

N(C) was the same string as C, so N's three clean losses are the model varying between calls. Although both of H's clean losses involved text Hazm had really changed, independent calls and denominators this small do not let us call that causal. Fourteen of 79 primary identical-input groups disagreed in label, and six in correctness. Every recovery exactly recreated C.

E. Earlier ZWNJ run (24 items)

The first ZWNJ run completed 432 of 432 requests. On its frozen 21-item primary set, request accuracy was 27/63 (42.9%) for C and P(C), 28/63 (44.4%) for N(C), 26/63 (41.3%) for N(P(C)) and 28/63 (44.4%) for both Hazm conditions. Majority accuracy was 9/21 everywhere except N(P(C)), at 8/21. Perturbation loss was 3/27 C-correct observations; N recovered 1/3 (non-exactly) and H recovered 3/3 (all exactly). Clean-input harm was 0/27 for both, and perturbed-input harm 2/27 for both.

All three losses came from one item, `peval-hard-reasoning-019`, whose stored answer was C ("two people"). Across three repeats, C answered C, C, C; P(C) answered A, A, A; N(P(C)) answered C, A, A; and H(P(C)) answered C, C, C. The perturbation replaced three instances of می‌دانند with می دانند. Hazm restored C exactly, and N did not. The item repeated across calls. That is an item-level observation and no evidence of a general ZWNJ effect.

Both perturbed-input harms in this run sit on `peval-challenge-ambiguity-016`. There N(P(C)) was text-identical to P(C), and H(P(C)) was text-identical to C and H(C), so the answer differences are the model varying between identical inputs. Four of 50 repeated identical-input groups disagreed in label, and three in correctness. The post hoc 20-item set drops reasoning-005, -007, -009 and -024; its request accuracies were 27/60 (45.0%) for C and P(C), 28/60 (46.7%) for N(C), 26/60 (43.3%) for N(P(C)), and 28/60 (46.7%) for both Hazm conditions, with loss and recovery counts unchanged.

F. A closer look at reasoning-019

This is a post hoc analysis. The item is a short set-overlap problem. With seven people, at least four French speakers and at least five English speakers, the smallest possible overlap is $4 + 5 - 7 = 2$, which is option C. Both of us confirmed that the clean and the perturbed wording keep this reasoning and the single correct answer. All three edits fall in the repeated verb می‌دانند. No number, quantifier, option or logical relation changes.

We tokenized both versions with the exact Qwen2.5 model file from the experiment. Both came to 76 tokens, so a longer prompt does not explain the effect. Each edited verb did change its local token boundaries, though: the ZWNJ-bearing and the space-bearing forms split into different token sequences, so the intervention swapped three ZWNJ-bearing sequences for three different space-bearing ones and left the token count alone.

One possible reading is that repeated local changes in representation unsettled a small quantized model on a multi-step arithmetic item. That is a hypothesis we formed after the run. We did not inspect internal activations, only one item showed the full pattern of C right and P(C) wrong, and identical inputs varied elsewhere. A larger preregistered study should test whether sensitivity grows with the number of altered ZWNJs, with verb-prefix edits, or with the number of reasoning steps.

G. Execution record and text-level checks

Table IX lists every stage of the project. Counts are shown by stage and not added up, since the stages share items and conditions. Every run was local and used no paid API.

For both perturbation families, all six conditions were generated before the corresponding model run. On the first 40 yeh/kaf items, H(C) and H(P(C)) were identical in all 40 pairs, but H(P(C)) equalled C in only 29, because Hazm also changed clean text; N(P(C)) equalled C in 37/40 with the spacing rules

TABLE VII
ACCURACY IN THE FIRST SIX-CONDITION YEH/KAF RUN.

Condition	Primary requests (n=105)	Primary majority (n=35)	Secondary requests (n=120)	Secondary majority (n=40)
C	53/105 (50.5%)	17/35 (48.6%)	54/120 (45.0%)	17/40 (42.5%)
P(C)	52/105 (49.5%)	18/35 (51.4%)	53/120 (44.2%)	18/40 (45.0%)
N(C)	52/105 (49.5%)	17/35 (48.6%)	53/120 (44.2%)	17/40 (42.5%)
N(P(C))	52/105 (49.5%)	17/35 (48.6%)	53/120 (44.2%)	17/40 (42.5%)
H(C)	52/105 (49.5%)	17/35 (48.6%)	55/120 (45.8%)	18/40 (45.0%)
H(P(C))	51/105 (48.6%)	17/35 (48.6%)	52/120 (43.3%)	17/40 (42.5%)

TABLE VIII
PAIRED METRICS IN THE FIRST SIX-CONDITION YEH/KAF RUN (PRIMARY SET).

Metric	N	H
Perturbation loss	6/53 (11.3%)	
Recovery among losses	5/6	5/6
Exact recovery	5/6	5/6
Non-exact recovery	0/6	0/6
Harm, clean input	3/53	2/53
Harm, perturbed input	5/52	6/52

TABLE IX
EXECUTION RECORD.

Stage	Items and conditions	Requests
Initial feasibility pilot	5 items × 4 cond.	20
Expanded one-repeat pilot	40 items × 4 cond.	160
Repeated yeh/kaf run	40 × 6 × 3	720
Repeated ZWNJ run	24 × 6 × 3	432
Yeh/kaf expansion	97 P/N-safe, 96 Hazm-safe items, 3 repeats	1,740
ZWNJ expansion	69 × 6 × 3	1,242
Targeted 7B replication	30 items × 2 × 3	180

on. On the 24 ZWNJ items, H(C) and H(P(C)) were identical in 20 pairs, Hazm restored C exactly in 18, and N in none, since N follows only its literal spacing rules.

H. What Hazm did to the clean questions

Hazm changed 11 of the 40 clean questions through 18 edit spans: 16 space-to-ZWNJ replacements and two deletions of fathatan (U+064B). After the run, both of us independently reviewed all 11 and agreed that meaning and the stored answer were preserved; that review is post hoc. Table X lists the changes; exact spans and longer notes are in `hazm_clean_change_audit.md`.

IX. TARGETED QWEN2.5-7B REPLICATION

The 1.5B model answers fewer than half of the questions correctly, so we asked whether a larger model behaves differently. We ran Qwen2.5-7B-Instruct-Q4_K_M (a two-part GGUF file) locally through llama.cpp. After a concurrency guard was added to the pipeline, the frozen targeted plan completed exactly 180 of 180 requests: 20 yeh/kaf pairs and 10 ZWNJ pairs, with C and P(C) only and three repeats. Every response was scorable, and none failed in transit. The 30 items were selected from 1.5B outcome strata before any 7B request was sent, so the set is not a random sample and cannot estimate how common discordance is in general.

Conditional discordance is counted over scorable C-correct requests, paired by item and repeat with P(C). On all 30 items, 23 of 43 C-correct requests from the 1.5B model became P(C)-incorrect, against 0 of 21 from the 7B model (Table XI). The 7B model, however, answered only 21 of 90 C requests correctly (23.3%), which leaves few observations in which a loss could occur at all. The absence of discordance is therefore inconclusive and cannot be attributed to greater robustness. We did not run a Q8_0 or Q5_K_M sensitivity check, because no corresponding 7B weight file is available locally, so we draw no conclusion about the effect of quantization. The complete machine-readable metrics are kept in `work/targeted_7b_replication_2026-10-21/clean_run/targeted_7b_metrics.json`.

X. DISCUSSION

Three observations stand out. First, the aggregate result and the item-level result point in different directions. Item-majority accuracy for C and P(C) was identical in both families, yet seven yeh/kaf items and two ZWNJ items got worse, and the same numbers got better. Flips of this size in both directions are as consistent with response variability as with sensitivity to spelling, and the repeated identical inputs show that variability is not small: 22 of 214 yeh/kaf groups and 16 of 174 ZWNJ groups disagreed in label. We did not test whether the perturbation effect exceeds this noise floor, so we make no claim either way.

Second, normalizer recoveries need careful reading. N restores C exactly by construction in the yeh/kaf family, so its 21 recoveries say little about the model. H recovered five yeh/kaf losses without restoring C; with denominators this small, and with identical inputs sometimes receiving different answers, those cases cannot be separated from chance.

Third, the 7B result limits what the cross-model comparison can show. A model that answers 21 of 90 C requests correctly (23.3%) offers few observations in which a loss could occur, so the absence of discordance says little about robustness. A real test needs a model, or an item set, on which baseline accuracy is more useful.

For people building Persian evaluation sets, three habits follow from the protocol rather than from our numbers: record the orthographic form of every question (code points and spacing), report normalizer results with exact restoration separated from answer recovery, and include repeated identical inputs as a baseline for response variability. These are methodological suggestions; our data do not show that they change accuracy.

TABLE X
HAZM CHANGES TO THE 11 CLEAN QUESTIONS IT ALTERED (ORIGINAL → HAZM). ALL WERE JUDGED TO PRESERVE MEANING AND THE STORED ANSWER.

Record	Hazm change	Category	Note
ambiguity-006	واقعا → واقعاً	diacritic removal	
ambiguity-008	گم کرده → گم‌کرده	verb spacing	
dev-culture-001	هفت سین → هفت‌سین; سین های → سین‌های	compound and suffix	
dev-culture-002	می شود → می‌شود	prefix verb	
dev-knowledge-002	ماه های → ماه‌های	suffix plural	
hard-culture-004	مناسب تر → مناسب‌تر	suffix comparative	
reasoning-001	three × می شود → می‌شود	verb prefix	
reasoning-002	می آید → می‌آید; محرمانه ای → محرمانه‌ای	suffix and verb prefix	Item already excluded for another defect.
reasoning-004	four verb-prefix spacing repairs	verb prefix	
reasoning-007	دقیقا → دقیقاً	diacritic removal	Item already marked ambiguous.
reasoning-010	شطرنج بازی → شطرنج‌بازی	compound and verb spacing	Confirmed explicitly by both of us.

TABLE XI
TARGETED REPLICATION ON 30 ITEMS, BY PERTURBATION FAMILY AND MODEL.

Family	Items	Model	C requests	P(C) requests	C majority	P(C) majority	C-correct → P(C)-incorrect
Yeh/kaf	20	1.5B	29/60 (48.3%)	28/60 (46.7%)	10/20 (50.0%)	10/20 (50.0%)	19/29 (65.5%)
		7B	15/60 (25.0%)	15/60 (25.0%)	5/20 (25.0%)	5/20 (25.0%)	0/15 (0%)
ZWNJ → space	10	1.5B	14/30 (46.7%)	15/30 (50.0%)	5/10 (50.0%)	5/10 (50.0%)	4/14 (28.6%)
		7B	6/30 (20.0%)	6/30 (20.0%)	2/10 (20.0%)	2/10 (20.0%)	0/6 (0%)
All	30	1.5B	43/90 (47.8%)	43/90 (47.8%)	15/30 (50.0%)	15/30 (50.0%)	23/43 (53.5%)
		7B	21/90 (23.3%)	21/90 (23.3%)	7/30 (23.3%)	7/30 (23.3%)	0/21 (0%)

XI. LIMITATIONS

Our literature set was supplied, not collected through a systematic review. Although a FarsEval-PKBETS PDF is now in the collection, the relation between that benchmark and our local files still needs verification, and the dataset metadata do not establish factual review. The preliminary screen was AI-assisted. The 117-item review and the transformation reviews were joint author adjudication rather than independent ratings, and heuristic near-duplicate detection missed two semantic duplicate pairs that we later found by hand. The 97 retained items are therefore author-approved, not externally validated, and no kappa is available.

The first two-author review came after the historical Version 7 run, so the filtered Version 7 analysis is post hoc. For Version 8, the same five exclusions were frozen before execution, which makes the 35-item yeh/kaf analysis prospective. The ZWNJ experiment preceded its detailed semantic review, so its 20-item reviewed set is post hoc.

Only the question body was transformed. N is an inverse-map positive control, not a general-purpose normalizer competing with Hazm on equal terms. Hazm 0.10.0 is incompatible with our Python 3.12 runtime, so execution used recorded dependency overrides. The 1.5B model is small and quantized, and its clean-condition majority accuracy was 46/97 (47.4%) for yeh/kaf and 32/69 (46.4%) for ZWNJ. The targeted 7B comparison covers 30 stratified items and should be read neither as a random replication nor as a prevalence estimate. The confidence intervals from the 1.5B expansion remain compatible with modest effects in either direction. Exact McNemar tests and item-clustered intervals handle the pairing; they do not resolve uncertainty from dataset selection, joint review, model scale, targeted sampling or identical-input variability.

XII. CONCLUSION

The study combines a complete 1.5B expansion (2,982 local requests on 97 yeh/kaf and 69 ZWNJ items) with a frozen, targeted 7B replication. In the 1.5B expansion, C and P(C) had identical item-majority accuracy in both families, and individual answers still changed under both perturbations. N's yeh/kaf recoveries were exact restoration by construction, while H produced both exact and non-exact ones; neither pattern shows a general accuracy benefit. On the 30 selected items, the 7B model had low absolute accuracy and no conditional discordance. That result is inconclusive, and the disappearance of discordance cannot be attributed to greater robustness.

The study provides a reproducible paired workflow and evidence that individual answers can flip. It cannot establish that spelling is irrelevant, that the conditions are statistically equivalent, or that Persian LLMs are robust in general. A larger, randomly sampled multi-model study with a quantization sensitivity check is still needed, followed by review from outside the author team.

Future work should (i) sample items randomly and use several models, including ones with higher baseline accuracy; (ii) run Q8_0 or Q5_K_M variants to separate model size from quantization; (iii) add reviewers from outside the author team;

(iv) test whether sensitivity grows with the number of altered ZWNJs; and (v) perturb the answer choices as well as the question body.

Data and Code Availability

The package contains this manuscript, the frozen pilot and expansion analysis plans, the aggregate yeh/kaf and ZWNJ metrics, the complete targeted 7B metrics, summary-level records of our review decisions, the Hazm change audit, the literature and novelty audits, the concurrency-guarded Python pipeline that prepares all six conditions, 26 passing tests, the pinned local library versions and the exact command lines for every run. Hazm is not bundled. Raw data, PDFs, model weights, runtime binaries, item-level prompts, raw responses and API keys are left out. The item-level review files and the run logs stay under `work/`, because redistribution permission for the data is unresolved.

Declarations

Author order, the affiliation shown above and co-corresponding authorship are confirmed. Funding, conflicts of interest, individual contributions and ethics applicability are still open. AI-assisted tools were used for drafting, code support and preliminary screening. Both authors reviewed the study materials, methods, analysis and final manuscript and remain responsible for the work. [Disclosure wording to be adjusted to the target venue's policy.]

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